
AI Shared Socioeconomic Pathways¹

Huw Hallam

Pablo Rosado
Our World in Data

Galina Lesnic
AI Safety Initiative
Groningen

With
Apart Research

Abstract

Climate change and AI governance face interconnected challenges through shared dependencies on energy, geopolitics, and cooperation. For both domains, the next 10-15 years may largely determine trajectories for the rest of the century. Yet forecasting efforts remain separate.

The IPCC Shared Socioeconomic Pathways (SSPs) offer a mature framework for scenario planning, integrating economic, demographic, technological, and governance factors through 2100. This project builds upon the SSPs to explore AI development risks for 2025-2040, mapping AI-specific outcomes onto each of the five baseline SSPs and examining how different constraints and policy choices could shift trajectories.

The aim is to bridge climate and AI safety communities by applying scenario planning methods from climate research to AI forecasting. Our scenarios are developed as a rapid proof-of-concept, not predictive forecasts or formal models. We provide an open-source implementation with interactive visualizations and illustrative projections based on IPCC data.

Keywords: AI forecasting, climate scenarios, Shared Socioeconomic Pathways, AI governance

¹ Research conducted at the AI Forecasting Hackathon, 2025,
<https://apartresearch.com/sprints/the-ai-forecasting-hackathon-2025-10-31-to-2025-11-02>

1. Introduction

AI development and climate change are two of humanity's most consequential challenges. For both, decisions in the coming decades may determine trajectories for the rest of the century and beyond. Both involve irreversible thresholds – climate tipping points and AI capability transitions that, once crossed, may be impossible to reverse.

Both domains share another critical feature: outcomes depend heavily on decisions by the same two major powers (the United States and China) who together dominate both global emissions and AI development.

If well-aligned, progress on both challenges could be mutually reinforcing: thoughtfully developed AI could accelerate climate solutions, while climate-conscious energy policy could shape AI development toward sustainability.

But, despite their deep interconnections, scenario planning for each domain proceeds independently. AI forecasts rarely account for climate and socioeconomic context, while climate scenarios do not yet model AI's transformative impacts.

The IPCC's Shared Socioeconomic Pathways (SSPs) (Riahi et al. 2017) offer a mature framework for modeling alternative futures through 2100, with broad scientific consensus. AI forecasting has produced valuable analyses and projections, though with far less consensus.

This project brings these two domains closer together by mapping AI development scenarios (2025-2040) onto the five baseline SSPs and in turn using these enriched pathways to bring clarity to AI development forecasting. To our knowledge, no prior work has systematically integrated AI forecasting with IPCC climate scenarios.

We acknowledge significant limitations: our scenarios rely heavily on qualitative judgment, and while we've sought to develop schematic models for our approach, our rapid exploration should be seen as an initial contribution to be developed further. Our goal is to demonstrate potential value in bridging these communities and inspire more thorough future work.

2. Methods

We map each SSP to AI development trajectories using qualitative reasoning and Our World in Data's IPCC Scenarios Data Explorer.²

For each scenario (baseline SSPs 1-5), we identify plausible positive and negative outcomes for AI development and societal impact, with our subjective estimation of each.

We focus on 2020 (present baseline), 2030, and 2040 to match IPCC decadal data availability. We take rough estimates for the AI-related metrics, our intention is that they give an initial 'best guess' of the risk profile of the path ahead and should be helpful for guiding policy until such time as they are able to be refined through further research.

We consider the following AI-related **metrics** (all on a scale 1-5):

- **Speed** (slow-fast) of AI development generally increases automation collapse, CBRN misuse, and existential risks unless offset by strong safety.
- **Cooperation** (unsafe-safe) reduces power-conflict and lock-in (transparency, shared norms).
- **Safety** (competitive-cooperative) measures, including regulation, reduces risk across the board, especially existential risks (evals, staged access, kill-switches and CBRN misuse).

² <https://ourworldindata.org/explorers/ipcc-scenarios>

We further assess each scenario by different AI-related **risk classes** (metrics are ordered by their relevance to the risk):

- **Economic collapse from rapid automation** (Speed, Safety, Cooperation)
- **AI-enabled power conflict** (Cooperation, Safety, Speed)
- **Permanent totalitarian lock-in** (Cooperation, Speed, Safety)
- **CBRN misuse** (Safety, Speed, Cooperation)
- **AI-caused existential catastrophe** (Safety, Speed, Cooperation)

Estimates of metrics and risk probabilities are educated guesses (reasoning provided in Appendices A and B).

Code and data: <https://github.com/pabloarosado/ai-ssp>

3. Results

We present five scenarios for AI development by 2040, mapping each to one of the IPCC’s baseline SSPs. We expect AI development to lead to faster socioeconomic transformation than climate change and indeed the physical climate impacts outlined by the IPCC for 2040 (e.g. warming, more frequent extreme weather events) are similar across their baseline SSPs in this near-term window. What differs is how societies prepare for and respond to those impacts—through infrastructure, governance, adaptation investment, and social support. We assume these elements of preparation will be broadly similar in the face of AI risk: if a cautious approach is taken to climate change and biodiversity risks, we expect a cautious approach to existential or economic risks from AI. This assumption of a similar risk tolerance may not always play out in reality, but by bridging the two policy approaches we hope to strengthen the likelihood of such a correspondence.

SSP1: Sustainability (Taking the Green Road)

Gradual, inclusive development prioritizing well-being, lower inequality, and efficient resource use.

AI development

- International cooperation on safety: shared testing, incident reporting, escalation rules.
- Cautious steps toward AGI, agreed red lines limit capabilities races.
- AI is framed as a global public good, benefits and access are broadly shared.
- Scaling aligned with energy, water, minerals, and emissions constraints.
- Managed automation: paced deployment and social cushions to protect livelihoods.

Possible variations

- Employment rates: augmentation vs. replacement requires strong transition policy to avoid unrest.
- Speed of development: clean-energy gains could re-accelerate AI beyond governance capacity.

SSP2: Middle of the Road

Historical trends continue with uneven national progress, slow movement toward sustainability, moderate population growth, and persistent inequality. Environmental degradation continues with occasional improvements.

AI development

- No global consensus on safety, patchwork national frameworks and different lab self-regulations.
- Competition and resource bottlenecks outweigh coordination.
- Policies react to shocks (e.g. labor, economic shocks) using standard redistribution measures (taxes, unemployment insurance, retraining).
- Ad-hoc reactive measures counter disinformation, epistemic erosion, and AI-enabled crime.
- AI goals often clash with sustainability targets, different countries strike different balances.

Possible variations

- Reactive stance increases vulnerability to unforeseen social/economic shocks.
- A salient “warning-shot” event may lead to stronger international coordination.

SSP3: Regional Rivalry (A Rocky Road)

Nationalism and security concerns drive countries inward, focus on regional self-sufficiency, slower growth, widening inequality, and worsening environmental degradation.

AI development

- Arms-race dynamics, cutting safety corners to gain advantage.
- AI wealth concentrates in model-owning and supply-chain nations, redistribution limited and uneven.
- Under-resourced governments mishandle unemployment, epistemic erosion, and AI-enabled crime, weakening institutions.
- Governmental AI spending focuses on defense, surveillance, and crime control.

Possible variations

- Socioeconomic fragility + extreme inequality raise risk of severe misuse (incl. CBRN).
- Militarization and misinformation increase odds of sliding toward deeper division/authoritarianism (toward SSP4).

SSP4 Inequality (A Road Divided)

Elites race ahead with advanced tech while most people are left out, eroding cohesion and raising conflict risk. Environmental policies favor high-income areas.

AI development

- Powerful models concentrated in elite hands to entrench control, safety oriented towards regime/asset protection.
- Mass access shaped by distraction, disinformation, and pervasive surveillance.
- Catastrophic-risk narratives may be leveraged for population control.
- Moves toward fortified “safe-havens” or off-planet relocations.

Possible variations

- Coercive dominance: elite uses AI to entrench control over large populations (repression/forced labor).
- Backlash & talent flight: resistance and engineer talent flight leading to slow capability growth.

SSP5 Fossil-fueled Development (Taking the Highway)

Rapid, market-driven growth with heavy fossil use, strong institutions but resource-intensive lifestyles and delayed mitigation.

AI development

- Corporate races with rapid iteration and compressed safety reviews (“fix it in the morning”).
- Energy/material bottlenecks, firms deploy tech to patch local environmental harms.
- Labor partially adapts, weak safety nets leave many displaced.
- Redistribution mainly via markets; high inequality persists.
- **High-variance path:** as likely to deliver breakthroughs as catastrophic failures.

4. Discussion and Conclusion

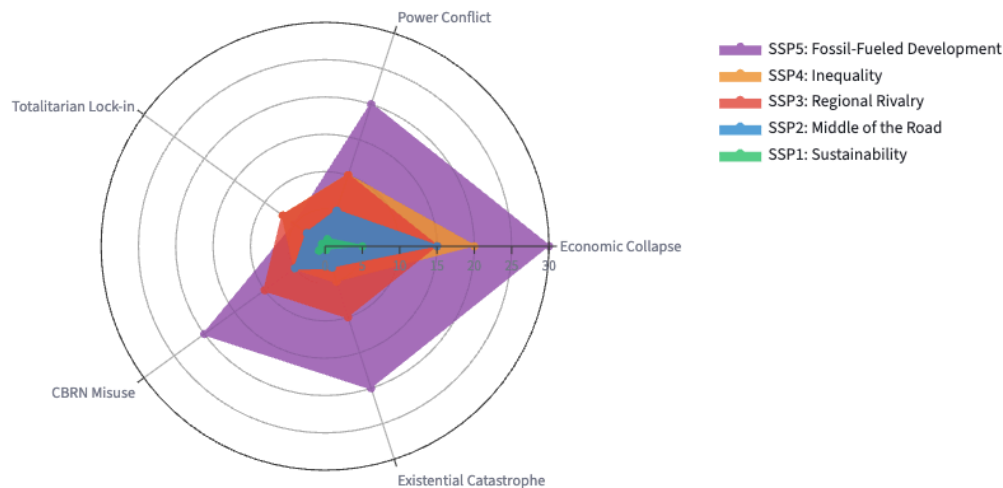
AI and climate futures are deeply interconnected through energy systems, geopolitical stability, and governance capacity. Yet scenario planning for each domain proceeds independently. By mapping AI development onto the IPCC’s SSP framework, this project has sought to illuminate the risk profiles of different approaches and to give a clearer picture of their outcomes in the next 15 years.

Although our findings are provisional, they give insight into how international cooperation and broader policy orientations may shape AI outcomes, affecting risk across multiple domains. We feel that this approach to

scenario mapping may help bring clarity to the current state of play in AI development, reducing ambiguity about future risk while showing alternative pathways that could lessen that risk.

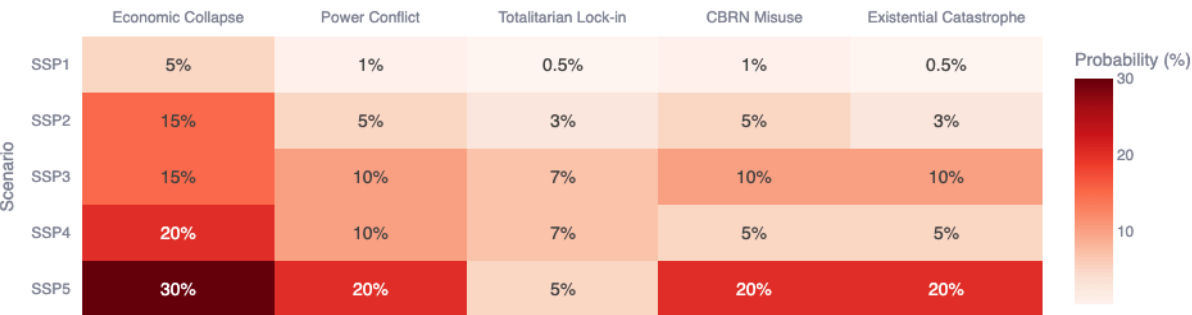
We hope our unique approach, linking AI safety and climate change mitigation, helps to bridge these two consequential policy fields. If well aligned, progress on both challenges may prove mutually reinforcing.

All Scenarios Compared



Scenario	Speed	Safety	Cooperation
SSP1: Sustainability	●●●○○	●●●●○	●●●●○
SSP2: Middle of the Road	●●●○○	●●○○○	●●○○○
SSP3: Regional Rivalry	●●○○○	●○○○○	●○○○○
SSP4: Inequality	●●●○○	●●○○○	●○○○○
SSP5: Fossil-Fueled Development	●●●●●	●○○○○	●●○○○

Risk assessment by scenario (probability by 2040)



5. References

Riahi et al. 2017. The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview, Global Environmental Change (2022).

1. Appendix A. AI-related Risks and Relevant Metrics

We consider the following AI-related risks (with metrics ordered by their relevance to risks):

- **Economic collapse from rapid automation:** Speed, Safety, Cooperation.

Faster automation shocks labor/product markets (Speed). Guardrails (Safety) pace deployment and add cushions (evals, phased rollout). Cooperation helps stability via shared best practices and coordinated policy, but is third-order.

- **AI-enabled power conflict:** Cooperation, Safety, Speed.

Misperception and escalation are mostly decreased by transparency, hotlines, and shared norms (Cooperation). Safety reduces accidental/unauthorized use. Speed adds pressure but, by itself, doesn't cause conflict without the cooperation failures.

- **Permanent totalitarian lock-in:** Cooperation, Speed, Safety.

Lock-in thrives with low transparency and high centralization (Cooperation). Speed concentrates power/compute before counter-institutions can respond. Safety helps, but classic alignment work doesn't automatically prevent political entrenchment.

- **CBRN-related misuse:** Safety, Speed, Cooperation.

Access controls, eval gates, staged releases (Safety) dominate. Speed raises leakage/exposure unless safety keeps pace. Cooperation matters for intel-sharing and harmonized rules, but is third.

- **AI-caused existential catastrophe:** Safety, Speed, Cooperation.

Alignment, oversight, and kill-switches (Safety) are first-order. Speed amplifies capabilities and compresses review time. Cooperation helps distribute audits/incident response globally.

2. Appendix B. Metrics & Risk Estimation for each SSP

SSP1

Safety: 4. Strong institutions, routine evaluations, staged access, and public-benefit R&D keep bad outcomes in check. It's not a 5 because meaningful innovation still proceeds and rules aren't so strict that they halt risky work entirely.

Cooperation: 4. Countries already coordinate on climate; that habit spills over into AI via shared benchmarks, incident reporting, and compatible standards. Not a 5 because some national interests remain and perfect alignment is rare.

Speed: 3. Energy access helps, but guardrails and staged rollout temper raw pace.

Economic collapse: 5%

Guardrails + paced rollout keep labor/product market shocks contained.

AI-enabled power conflict: 1%

High cooperation + institutions decrease escalation risk.

Permanent totalitarian lock-in: 0.5%

Inclusive governance + checks & balances limit lock-in possibilities.

CBRN-related misuse: 1%

Strong safety culture and staged access reduce dual-use leakage.

AI-caused existential catastrophe: 0.5%

High safety and coordination keep catastrophic pathways rare.

SSP2

Safety: 2. Policy is mostly reactive. Safety practices exist but are uneven and often tighten only after incidents.

Cooperation: 2. Rules and norms vary by region, leading to frictions and weak sharing.

Speed: 3. Fast where capital & chips exist, slow elsewhere (net “medium”, high variance).

Economic collapse: 15%

Uneven adoption + reactive policy allow faster shocks in some regions.

Power conflict: 5%

Different regional standards create misperceptions and destabilizing incidents.

Lock-in: 3%

Lack of coordinated accountability might lead to some concentration.

CBRN misuse: 5%

Mixed access controls and gaps in evals/reviews lead to some leakage risk.

Existential catastrophe: 3%

Moderate speed with weak, uneven safety moderately increases unlikely risks.

SSP3

Safety: 1. “Security theater” replaces true systemic safety. Secrecy, rushing to outpace rivals, and weak reviews let accidental risks accumulate.

Cooperation: 1. Minimal cross-border sharing, standards don’t align, incidents are hidden.

Speed: 2. Export controls on materials/energy drag the average down.

Economic collapse: 15%

Slower civilian speed but brittle, shock-prone economies suffering from export controls.

Power conflict: 10%

Arms race + low transparency increase escalation risk.

Lock-in: 7%

Security-centric governance with few accountability channels and low cooperation.

CBRN misuse: 10%

Fragmented oversight + incentives for secrecy increase misuse risk.

Existential catastrophe: 10%

Secrecy undermines cross-checking and incident response, even if the speed is moderately low.

SSP4

Safety: 2. Elite enclaves protect their own assets well, but harms to others aren’t mitigated, lowering overall safety

Cooperation: 1. Minimal cross-border sharing, incidents are hidden.

Speed: 3. Enclaves move quickly, while the majority lags, so average speed with high variance

Economic collapse: 20%

Fast in enclaves + social fragility elsewhere leads to higher systemic fragility and shock risk.

Power conflict: 10%

Grievances + asymmetric capabilities increase escalation/rebellion risk.

Lock-in: 7%

Surveillance/profiling + concentrated compute can entrench regimes.

CBRN misuse: 5%

Speed and labs are concentrated in enclaves, so possible leakage from peripheries.

Existential catastrophe: 5%

Concentrated capability + weak oversight leads to some tail risks.

SSP5

Safety: 1. Market concerns dominate, weak reviews, minimal guardrails.

Cooperation: 2. Some industry cooperation exists, but competition rules.

Speed: 5. Capital + energy + permissive policy = unconstrained scaling.

Economic collapse: 30%

Very fast automation + compressed review leads to a large chance of tech-driven recession by 2040.

Power conflict: 20%

Rapid capability gains + competitive geopolitics increase crisis/accident probability.

Lock-in: 5%

Centralization + surveillance increase entrenchment risk, but there's still some cooperation.

CBRN misuse: 20%

Fast scaling + weak guardrails increase dual-use leakage probability.

Existential catastrophe: 20%

High speed with weak safety increases alignment/accident tail risks.